

## BY 530: Advanced Cell and Molecular Biology

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| Course Code   | BY 530                                                                                |
| Course Name   | Advanced Cell and Molecular Biology                                                   |
| L-T-P-C       | 3-0-0-3                                                                               |
| Prerequisites | IC136 - Understanding Biotechnology & its Applications or Consent of faculty member   |
| Intended for  | Core for M.Tech Biotechnology and elective for B.Tech, M.Sc/M.Tech and PhD candidates |
| Approval      | 42nd Senate (Jan 2024), in the M.Tech Biotechnology curriculum; 54th BoA (May 2024)   |

### Course Contents

- **INTRODUCTION:** A tour to Cell: Exploring Eukaryotic, animal and plant cells; cell compartments, their structure and function. Introduction to DNA, RNA and Proteins. Replication and hereditary. Epigenetics, transcription, and translation and their regulatory mechanisms. (6 hours)
- **CELL CYCLE:** Eukaryotic cell cycle: Biochemical and genetics studies on cell cycle; mechanisms regulating mitotic events; meiosis - a special type of cell division; cell birth, lineage and death: Asymmetrical cell division, patterns of stem cell division; Biological description of apoptosis; Molecular basis of cancer, oncogenes and tumor suppressor genes. (4 hours)
- **CELL SIGNALLING, MEMBRANE TRANSPORT and TRAFFICKING:** Membrane structure, function and models; evolution of different membrane lipids, Membrane proteins and their functions, Membrane carbohydrates and their roles in cell-cell recognition; The permeability of the bi-layers: transport proteins; Passive, active and co transport by antiporters and symporters, transporter proteins in plant vacuoles exocytosis and endocytosis Cell signalling and transport across cell membranes: Signalling molecules and cell surface receptors; intracellular signal transduction; G protein coupled receptors; plant growth factors and hormones-auxins, gibberlines, cytokines and others passive and active transport; transport into prokaryotic cells; endocytosis, exocytosis; entry of viruses and toxins into cells. Membrane trafficking: Translocation of secretory proteins across the ER membrane; protein modifications, folding and quality control in the ER; export and sorting of proteins to mitochondria, chloroplast and peroxisomes. (12 hours)
- **GENE MANIPULATION AND ITS APPLICATIONS:** A tutorial on Software and online/freeware tools for analyzing DNA sequence and virtual cloning. PCR and variations of PCR. DNA manipulative enzymes (polymerases, DNA modifying enzymes, nucleases, ligases, etc.) and their molecular biology applications. Different kinds of cloning vectors, cloning and expression vectors, (bacterial, plant and animal vectors, viral vectors etc.). Cloning techniques: Traditional (restriction digestion/ligation) cloning and its variations with use of linkers and adaptors. PCR cloning, multi fragment cloning techniques (Golden gate assembly, infusion cloning, Gibson assembly, etc.). Techniques for selection and screening of clones. Methods for gene and genome manipulation (eg. RNAi, CRISPR etc) Applications of Genetic Engineering in Biotechnology (Therapeutic hormones/proteins, protein and RNA vaccines, Synthetic food, GM plants etc.). (12 hours)
- **MOLECULAR TECHNIQUES FOR GENE EXPRESSION ANALYSIS:** Introduction to chromatin organization and regulation. Techniques to study replication and transcription (eg. replication labeling, NGS, ChIP, ATACseq, 3D chromatin techniques, etc.), and translation (eg. reporter assays, Gel shift/EMSA, ribosome profiling etc.). (8 hours)

### Text Books

1. Lodish et al., **Molecular Cell Biology**, 5th Edition or Recent, W.H. Freeman and Company.
2. Alberts et al., **Molecular Biology of the Cell**, 4th Edition or Later, Garland Sciences.

### References

1. NA